

Joint TR Beam Allocation and Multidimensional Resource Optimization Scheme for Multi-Target Tracking in Airborne Radar Networks

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Abstract—In this paper, we propose a joint transmit-receive (TR) beam allocation and multidimensional resource optimization (JTBA-MDRO) scheme for multi-target tracking (MTT) in airborne radar networks (ARNs). Joint resource optimization is achieved by formulating and solving an optimization problem that aims to minimize total power consumption while satisfying the predefined MTT accuracy requirements for each target. An analytical expression for the adaptive detection threshold is derived, which enables adaptive adjustment of detection thresholds for ARNs. Based on the above thresholds, the mathematical expressions of target detection probability and Bayesian Cramér-Rao lower bound (BCRLB) are derived as the evaluation metrics for MTT, respectively. The underlying optimization problem is nonconvex and nonlinear, which is solved through a four-step decomposition algorithm. Simulation results verify the effectiveness and superiority of the presented JTBA-MDRO scheme.

Index Terms—Airborne Radar Networks, Beam Allocation, Resource Optimization, Multi-target Tracking.

I. INTRODUCTION

Unmanned aerial vehicles (UAVs) equipped with airborne radars are crucial for surveillance tasks, including target tracking [1]. The mobility of airborne radar networks (ARNs) allows dynamic optimization of spatial spread and resource, enhancing system performance beyond traditional fixed networks [2]. However, the radio frequency (RF) resource of airborne radars is limited by the weight and size constraints of the platform [3]. Additionally, the low probability of intercept (LPI) is essential in modern battlefield applications [4]. Therefore, enhancing the LPI performance in ARNs while maintaining multi-target tracking (MTT) performance has become a key research focus.

Consequently, resource management in ARNs has attracted significant attention. Lu et al. [5] proposed an LPI-based resource allocation strategy to track targets in moving ARNs in complex electromagnetic environments. Reference [6] designed a multidomain resource allocation algorithm for heterogeneous multiple radar networks under non-ideal detection conditions for asynchronous MTT scenarios. Reference [7] developed a joint beamforming and power control game algorithm for MIMO radar in multiple jammer environments.

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Lin et al. [8] proposed a beam allocation and RF resource management algorithm for network radar to address insufficient target resolution in group target tracking. Guo et al. [9] proposed a joint transmit-receive (TR) beam selection and power allocation strategy for MTT in netted radar systems, in order to maximize target tracking performance under limited radar system resource.

The aforementioned research provides a solid foundation for resource allocation techniques in ARNs, but certain limitations still exist. The existing methods primarily focus on improving the tracking accuracy in ARNs but fail to adequately enhance LPI performance. The existing resource optimization methods do not account for the adaptive adjustment of the detection thresholds.

To address these problems, this paper proposes a joint transmit-receive beam allocation and multidimensional resource optimization (JTBA-MDRO) scheme, which aims to minimize total power consumption while enhancing the LPI performance of ARNs, subject to given detection performance constraints.

II. SYSTEM MODEL

A. Problem Scenario

Consider a scenario in which ARNs with M UAVs perform MTT. Each UAV carries a phased array airborne radar capable of simultaneously transmitting and receiving multiple beams. There are N ($N \geq M$) moving targets to be tracked.

Assuming constant velocity (CV) motion for both targets and UAVs [10], with the sampling interval of T , the dynamic model of each target is:

$$\mathbf{X}_{n,k}^{\text{tar}} = \mathbf{F}_n^{\text{tar}} \mathbf{X}_{n,k-1}^{\text{tar}} + \mathbf{N}_n \quad (1)$$

where $\mathbf{X}_{n,k}^{\text{tar}}$ and $\mathbf{X}_{n,k-1}^{\text{tar}}$ denote the state vectors of the n th target at time k and time $k-1$, respectively; $\mathbf{F}_n^{\text{tar}}$ denotes the transition matrix of target state; $\mathbf{N}_n$ denotes the zero-mean Gaussian process noise with covariance $\mathbf{C}_n$.

B. Radar Measurement Model

Let $\mathbf{z}_{m,n,k}^q$ denote the q th state measurement of target n by airborne radar m at time k .

$$\mathbf{z}_{m,n,k}^q = \begin{cases} \mathbf{h}_m(\mathbf{X}_{m,k}^{\text{tar}}) + \mathbf{Y}_{m,n,k}, & \text{Target} \\ \mathbf{u}_{m,n,k}^{\text{false}}, & \text{False alarm} \end{cases} \quad (2)$$

where $\mathbf{u}_{m,n,k}^{\text{false}}$ represents the false alarm, $\mathbf{Y}_{m,n,k}$ represents the zero-mean Gaussian measurement noise, and $\mathbf{h}_m(\cdot)$ represents the nonlinear measurement function.

Considering the impact of transmit-receive separation in ARNs, the echo signal-to-noise ratio (SNR) can be expressed as

$$\text{SNR}_{m,n,k} = \frac{T_0}{T_1} \frac{P_{m,n,k} G_t G_r \lambda^2 \text{RCS}_{m,n,k}}{(4\pi)^3 k_B T_e B_{m,n,k}^r F_r (R_{i,n,k}^{\text{tr}} R_{j,n,k}^{\text{re}})^2} \quad (3)$$

where T_0 denotes the radar dwell time, T_1 denotes the pulse repetition interval, $P_{m,n,k}$ denotes the transmit power of radar m to target n at time k , G_t and G_r denote radar transmit antenna gain and radar receiving antenna gain, respectively, λ denotes the radar signal wavelength, $\text{RCS}_{m,n,k}$ denotes the radar cross section (RCS) of target n at time k , k_B denotes the Boltzmann constant, T_e denotes the temperature, $B_{m,n,k}^r$ denotes the matched filter bandwidth, F_r denotes the receiver noise figure, $R_{i,n,k}^{\text{tr}}$ denotes the distance from transmit node i to target n , and $R_{j,n,k}^{\text{re}}$ denotes the distance from receive node j to target n .

III. PROPOSED JTBA-MDRO SCHEME

A. MTT Performance

Based on hypotheses H_1 (target present) and H_0 (target absent), the pre-detection threshold of the airborne radar m at time k for target n located within the l -th resolution cell can be expressed as

$$\gamma_{m,n,k}^l = A_{m,n,k}^l \geq \frac{1 + \text{SNR}_{m,n,k}}{\text{SNR}_{m,n,k}} \cdot \ln \left[\frac{(1 + \text{SNR}_{m,n,k}) \eta_{m,n,k}}{\rho(\mathbf{z}_{m,n,k}^q; \mathbf{z}_{m,n,k|k-1}, \mathbf{E}_{m,n,k|k-1})} \right] \quad (4)$$

where $\gamma_{m,n,k}^l$ represents the echo detection threshold, $A_{m,n,k}^l$ represents the detected echo amplitude, $\rho(\cdot)$ represents a joint Gaussian distribution with mean $\mathbf{z}_{m,n,k|k-1}$ and covariance $\mathbf{E}_{m,n,k|k-1}$, and $\eta_{m,n,k}$ represents the pre-detection threshold.

The average detection probability $P_{Dm,n,k}^{\text{av}}$ is expressed as

$$P_{Dm,n,k}^{\text{av}} \approx [(1 + \text{SNR}_{m,n,k}) \eta_{m,n,k}]^{-\frac{1}{\text{SNR}_{m,n,k}}} \cdot \xi_{m,n,k}^{-\frac{\xi_{m,n,k}}{2}} \cdot \left[\frac{\sqrt{\xi_{m,n,k}}}{\sqrt{(2\pi)^\omega \|\mathbf{E}_{m,n,k|k-1}\|}} \right]^{\xi_{m,n,k}-1} \quad (5)$$

where $\xi_{m,n,k} = 1/\text{SNR}_{m,n,k} + 1$, and ω denotes the measurement dimension. It can be seen that $P_{Dm,n,k}^{\text{av}}$ is related to $\text{SNR}_{m,n,k}$, coupling with the pre-detection threshold, transmit power, and signal bandwidth.

The Bayesian information matrix (BIM) for the predicted state can be used to describe MTT performance [11]. We introduce the information decay factor $\kappa_m(\mathbf{X}_{n,k}^{\text{tar}}) \in [0, 1]$ to characterize the impact of measurement source uncertainty on target tracking performance [12], [13]. The BIM can be given by

$$\mathbf{J}(\mathbf{X}_{n,k|k-1}^{\text{tar}}) = [\mathbf{C}_n + \mathbf{F}_n^{\text{tar}} \mathbf{J}^{-1}(\mathbf{X}_{n,k-1}^{\text{tar}}) \mathbf{F}_n^{\text{tar}}]^{-1} + \sum_{m=1}^M \mathbb{E} \left[\kappa_m(\mathbf{X}_{n,k}^{\text{tar}}) \mathbf{H}_{m,n,k|k-1}^T \mathbf{G}_{m,n,k|k-1}^{-1} \mathbf{H}_{m,n,k|k-1} \right] \quad (6)$$

where $\mathbb{E}[\cdot]$ denotes expectation, $\mathbf{G}_{m,n,k|k-1}$ denotes the predicted measurement noise covariance, $\mathbf{H}_{m,n,k|k-1}$ refers to the Jacobian matrix of $\mathbf{h}_m(\mathbf{X}_{n,k-1}^{\text{tar}})$.

The predicted Bayesian Cramér-Rao lower bound (BCRLB) matrix is the inverse of BIM

$$\mathbb{C}_{n,k|k-1} = [\mathbf{J}(\mathbf{X}_{n,k-1}^{\text{tar}})]^{-1} \quad (7)$$

On this basis, the tracking accuracy of target n at time k could be expressed as

$$\mathbb{R}_{n,k|k-1} = \sqrt{\mathbb{C}_{n,k|k-1}(1,1) + \mathbb{C}_{n,k|k-1}(3,3) + \mathbb{C}_{n,k|k-1}(5,5)} \quad (8)$$

where $\mathbb{C}_{n,k|k-1}(1,1)$, $\mathbb{C}_{n,k|k-1}(3,3)$ and $\mathbb{C}_{n,k|k-1}(5,5)$ represent the elements in the matrix that correspond to the accuracy of the target range estimation.

Therefore, the MTT accuracy function is defined as

$$\mathbb{F}_{k|k-1}(u_{m,n,k}^{\text{tr}}, u_{m,n,k}^{\text{re}}, \eta_{m,n,k}, \text{SNR}_{m,n,k}) = \sum_{n=1}^N \mathbb{R}_{n,k|k-1} \quad (9)$$

where $u_{m,n,k}^{\text{tr}}$ and $u_{m,n,k}^{\text{re}}$ represent the transmit and receive beam allocation parameters, respectively. $\eta_{m,n,k}$ represents the pre-detection threshold. $\text{SNR}_{m,n,k}$ depends on the transmit power $P_{m,n,k}$, the signal bandwidth $B_{m,n,k}$, and the position of UAV.

B. Problem Formulation

The objective of the proposed JTBA-MDRO strategy is to minimize the total power consumption of ARNs while satisfying the specified thresholds of the target detection probability, MTT accuracy, and RF resource constraints. Based on this, the JTBA-MDRO model can be formulated as follows:

$$\begin{aligned} & \min_{u_{m,n,k}^{\text{tr}}, u_{m,n,k}^{\text{re}}, \eta_{m,n,k}, P_{m,n,k}, B_{m,n,k}, \theta_{m,k}, \psi_{m,k}} \sum_{m=1}^M \sum_{n=1}^N P_{m,n,k} \\ & \text{s.t.} \quad \begin{cases} P_{D\min} \leq P_{Dm,n,k}^{\text{av}} < 1, \mathbb{F}_{k|k-1} \leq \mathbb{F}_{\max}, & \forall m, n, k \\ \sum_{m=1}^M u_{m,n,k}^{\text{tr}} \leq U_{\text{tr}}, \sum_{n=1}^N u_{m,n,k}^{\text{tr}} \leq L, & \forall m, n, k \\ \sum_{m=1}^M u_{m,n,k}^{\text{re}} \leq U_{\text{re}}, & \forall m, n, k \\ \eta_{\min} \leq \eta_{m,n,k} \leq \eta_{\max}, & \forall m, n, k \\ P_{\min} \leq P_{m,n,k} \leq P_{\max}, B_{\min} \leq B_{m,n,k} \leq B_{\max}, & \forall m, n, k \\ \sum_{n=1}^N P_{m,n,k} \leq P_w, \sum_{n=1}^N B_{m,n,k} \leq B_w, & \forall m, n, k \\ \theta_{\min} \leq \theta_{m,k} \leq \theta_{\max}, \psi_{\min} \leq \psi_{m,k} \leq \psi_{\max}, & \forall m, k \\ u_{m,n,k}^{\text{tr}}, u_{m,n,k}^{\text{re}} \in \{0, 1\}, & \forall m, n, k \end{cases} \end{aligned} \quad (10)$$

where P_{Dmin} and $\mathbb{F}_{\text{max}}$ denote the minimum average target detection probability threshold and the maximum target tracking accuracy threshold, respectively. L represents the upper limit on the number of beams that an individual airborne radar node can allocate to illuminate a single target. η_{max} and η_{min} indicate the maximum and minimum values of the airborne radar's pre-detection threshold, respectively. P_{max} and P_{min} denote the maximum and minimum allowable transmit power values, respectively. B_{max} and B_{min} specify the maximum and minimum allowable signal bandwidth values, respectively. P_w represents the total transmit power upper limit for an individual airborne radar node and B_w denotes the total signal bandwidth upper limit for an individual airborne radar node.

C. Solution Technique

As seen in problem (10), the above optimization problem is a nonconvex, nonlinear problem with multiple performance constraints. We introduce a four-step decomposition algorithm, which is solved stepwise using semi-definite programming (SDP), greedy algorithm, tabu search, and sequential quadratic programming (SQP). The specific steps are as follows:

Step 1: Transmit beam allocation. Fix the positions of the UAVs and the RF resource of ARNs, assuming that the receive beam allocation is identical to the transmit beam allocation. We use the SDP algorithm to relax this problem into a convex optimization problem. Using the properties of the Schur complement theorem, an auxiliary matrix $\mathbf{A}_n$ is introduced to replace the objective function, with the following constraints added:

$$\mathbf{A}_n \geq \mathbb{C}_{n,k|k-1} \Rightarrow \begin{bmatrix} \mathbf{A}_n & \mathbf{I}_4 \\ \mathbf{I}_4 & \mathbf{J}(\mathbf{X}_{n,k-1}^{\text{tar}}) \end{bmatrix} \geq \mathbf{0} \quad (11)$$

Compared to other algorithms, the interior-point method can incorporate various system constraints relatively easily and can solve problems in polynomial time. Therefore, we use the interior-point method to solve the relaxed convex optimization problem and obtain the transmission beam weights. Under the given constraints, the beam with the highest weight is selected for the illumination of the target, thus finalizing the allocation of the transmit beam $u_{m,n,k}^{\text{tr}}$.

Step 2: Receive beam allocation. Based on the results of Step 1, we number the echoes reflected from each target sequentially. Receive beam allocation is a large-scale discrete optimization problem; greedy algorithms are highly computationally efficient and always converge to a feasible local optimum. Thus, we sequentially consider the allocation of receiving nodes for these echoes through a greedy algorithm, the final receive beam allocation $u_{m,n,k}^{\text{re}}$ is obtained.

Step 3: RF resource optimization. The variables in RF resource optimization are coupled with the constraints, and the joint optimization involves a large number of parameters, making it a nonlinear, nonconvex, high-dimensional multivariate optimization problem. Although tabu search is a heuristic method and cannot guarantee a strict optimal solution, its tabu table mechanism effectively prevents getting stuck in local minima and ensures stable convergence to high-quality

solutions within a finite number of iterations. With beam allocation fixed, the pre-detection thresholds, transmit power, and signal bandwidth are optimized via tabu search to enhance the LPI performance of ARNs.

Step 4: UAV trajectory planning. The SQP algorithm leverages the derivative information of the objective and constraint functions to achieve efficient convergence. While UAVs follow a CV motion model, we use SQP to optimize the pitch angle $\theta_{m,k}$ and the heading angle $\psi_{m,k}$ for trajectory planning in ARNs.

IV. SIMULATION RESULTS

The simulations in this section were performed using MATLAB R2024a. The measurement noise was modeled as zero-mean Gaussian white noise. The RCS fluctuation of the target followed the Swerling Type I model, and the process noise intensity was set to $\sigma_q^2 = 10$. All simulation results were averaged over 100 Monte Carlo runs.

A scenario in which ARNs composed of $M = 3$ UAVs detected and tracked $N = 4$ moving targets was considered. Each radar node could generate up to $U_{\text{tr}} = 3$ transmit beams and receive up to $U_{\text{re}} = 4$ beams simultaneously. The sampling interval was $\Delta T = 3$ s and a single target could be illuminated by at most $L = 2$ beams. Table I presents the specific values of each parameter used in the simulations.

TABLE I: Parameter values used in the simulations

Parameter symbol	Value	Parameter symbol	Value
P_{max}	9000 W	P_{min}	900 W
B_{max}	8 MHz	B_{min}	0.08 MHz
η_{max}	6	η_{min}	0.02
P_w	22500 W	B_w	20 MHz
$\mathbb{F}_{\text{max}}$	250 m	$P_{\text{D min}}$	0.9

The trajectories of airborne radars and multi-targets are shown in Fig. 1. It shows that UAVs adjust their trajectories toward targets over time to improve tracking accuracy.

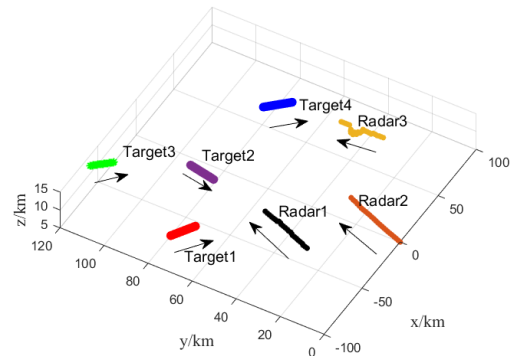


Fig. 1: Trajectories of airborne radars and multiple targets.

The results of the transmit beam allocation and resource optimization are shown in Fig. 2. It can be seen that target 1 is mainly illuminated by radar 1 and radar 2 due to the closest proximity, while target 3 is mainly illuminated by radar

1 and radar 3. Furthermore, ARNs allocate a large amount of RF resource to farther targets to maintain better tracking performance. For example, target 3 remains farther from the radar than the other targets throughout the simulation, so it is allocated the most RF resource.



Fig. 2: Transmit beam and resource allocation results.

The results of the receive beam allocation are shown in Fig. 3. Radars tend to receive echoes from targets that are closer to them. Since radar 1 is closer to the targets than the other radars, it receives the most echoes. Furthermore, radar 1 primarily receives two echoes from target 1, while radar 3 primarily receives two echoes from target 4.

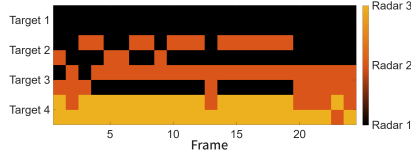


Fig. 3: Receive beam allocation results.

To further highlight the reliability of the proposed scheme in optimization effectiveness, it is compared with three benchmark schemes while keeping other parameters unchanged:

(1) Benchmark 1: It fixes the pre-detection thresholds, uses tabu search to allocate transmit power and signal bandwidth, and uses SDP to optimize UAV trajectories.

(2) Benchmark 2: It uses SDP to sequentially optimize the pre-detection thresholds, transmit power, signal bandwidth, and UAV trajectories.

(3) Benchmark 3: It fixes the UAV trajectories and uses the proposed scheme to optimize the pre-detection thresholds, transmit power, and signal bandwidth.

Fig. 4 and Fig. 5 show the average root mean square error (ARMSE) of tracking each target and total power consumption under different schemes, respectively. The proposed scheme achieves superior tracking accuracy and lower total power consumption. The total power consumption and ARMSE for benchmark 1 and benchmark 2 are relatively high. For benchmark 3, as airborne radar nodes are far from targets, the corresponding echo SNR is reduced, leading to lower tracking accuracy; but since its RF resource optimization steps are the same as the proposed scheme, their total power consumption is similar.

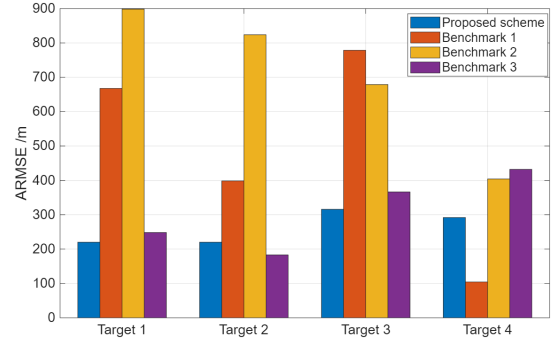


Fig. 4: Comparison of the target tracking ARMSE results.

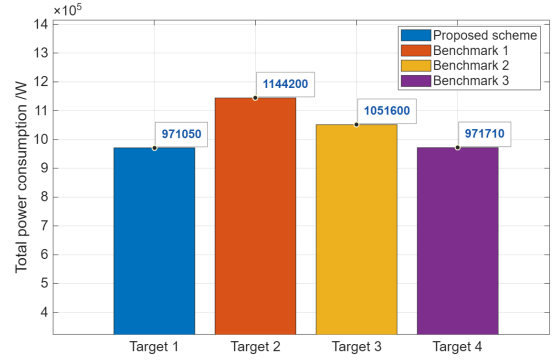


Fig. 5: Comparison of the total power consumption results.

V. CONCLUSION

This paper proposed a JTBA-MDRO scheme for MTT in ARNs. By jointly optimizing beam allocation, detection thresholds, transmit power, bandwidth, and UAV trajectories, the scheme minimizes the RF resource while satisfying tracking accuracy constraints. Simulation results show that, in comparison with existing schemes, the proposed scheme can effectively reduce the total power consumption of ARNs, enable efficient joint optimization over transmit-receive beam and multidimensional resource, and thus enhance the LPI capability of ARNs.

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